

Medical Image Similarity Explorer

Project Phase: Prototype Investigation

Target Dataset: BreastMNIST (Medical Imaging)

Core Model: ResNet-18 (Transfer Learning & Fine-Tuning)

Status: Milestone 1 Completed

Project Context & Motivation

This prototype investigates how similarity between medical images changes depending on whether similarity is measured directly in raw pixel space or within a learned feature space. This work serves as the foundational technical backbone for our upcoming interactive Medical Image Similarity Explorer.

Initial Challenges & Empirical Findings

Our initial pipeline utilized a standard ResNet-18 model (`torchvision.models.ResNet18_Weights.DEFAULT`) pretrained exclusively on natural RGB images (ImageNet), but lacking medical domain pretraining.

During our first attempts without model tuning, the distribution visualizations yielded irrelevant data showing virtually no difference between similarity scores for same-label versus different-label pairs. While this overlapping behavior was understandable for raw pixel similarity (due to background noise and low-level intensity overlap), it was unexpected and problematic for feature-space similarities.

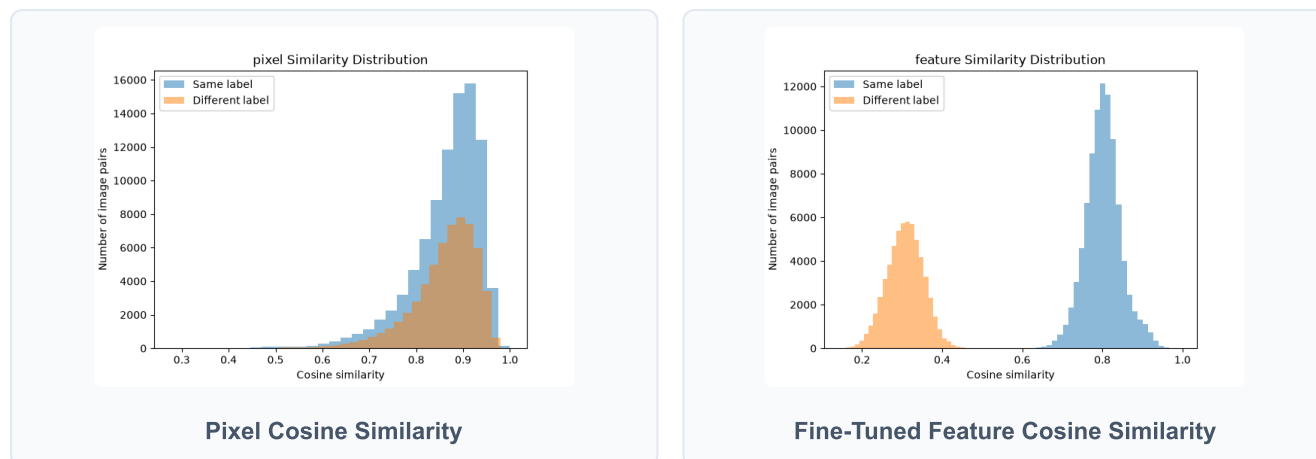
Key Insight: Off-the-shelf ImageNet representations focus heavily on natural color textures and macro-objects, which fail to capture the clinical morphology required to differentiate pathology labels in biomedical scans like BreastMNIST.

Methodological Solution

To overcome this barrier, we fine-tuned the ResNet-18 model using Gemini-guided implementation protocols directly on the BreastMNIST dataset. We trained the network for 10 epochs—just enough to observe a crucial, sharp drop in training loss. This adapted the backbone's latent space from generic natural features to specialized medical patterns.

Comparative Evaluation & Results

After the 10-epoch fine-tuning process, we stripped the classification head and fed the adapted feature embeddings back into our pairwise cosine similarity algorithm. The resulting distribution plots conclusively demonstrate the power of domain adaptation over raw pixel analysis.



Analysis of Results

Pixel Similarity: As highlighted by the overlapping curves on the left, pixel-level cosine similarity fails to distinguish between same-label and different-label pairs, clustering densely between 0.85 and 0.95 due to high-dimensional proximity bias.

Fine-Tuned Feature Similarity: In sharp contrast, the fine-tuned feature space (right plot) achieves clean separation. Same-label pairs concentrate at a significantly higher cosine similarity (~ 0.80 – 0.85), whereas different-label pairs shift downwards to a distinct distribution centered around ~ 0.33 .